

Latent Conditional Rectified Flow for MRI Synthesis Across Modalities and Field Strengths in the MRlxFIELDS2026 Challenge

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MOTIVATION

MRI imaging is central to clinical care, but differences in field strength create major variations in contrast, resolution, and noise, limiting data comparability across hospitals and multi-centre studies. Image synthesis could address this challenge by enabling consistent translation across field strengths.

PROPOSAL

LCRFM is a 3D latent diffusion model that translates brain MRI across three modalities (T1, T2, FLAIR) and five field strengths (0.1T–7T). It uses rectified flow in a compact latent space with anatomical guidance [1,2].

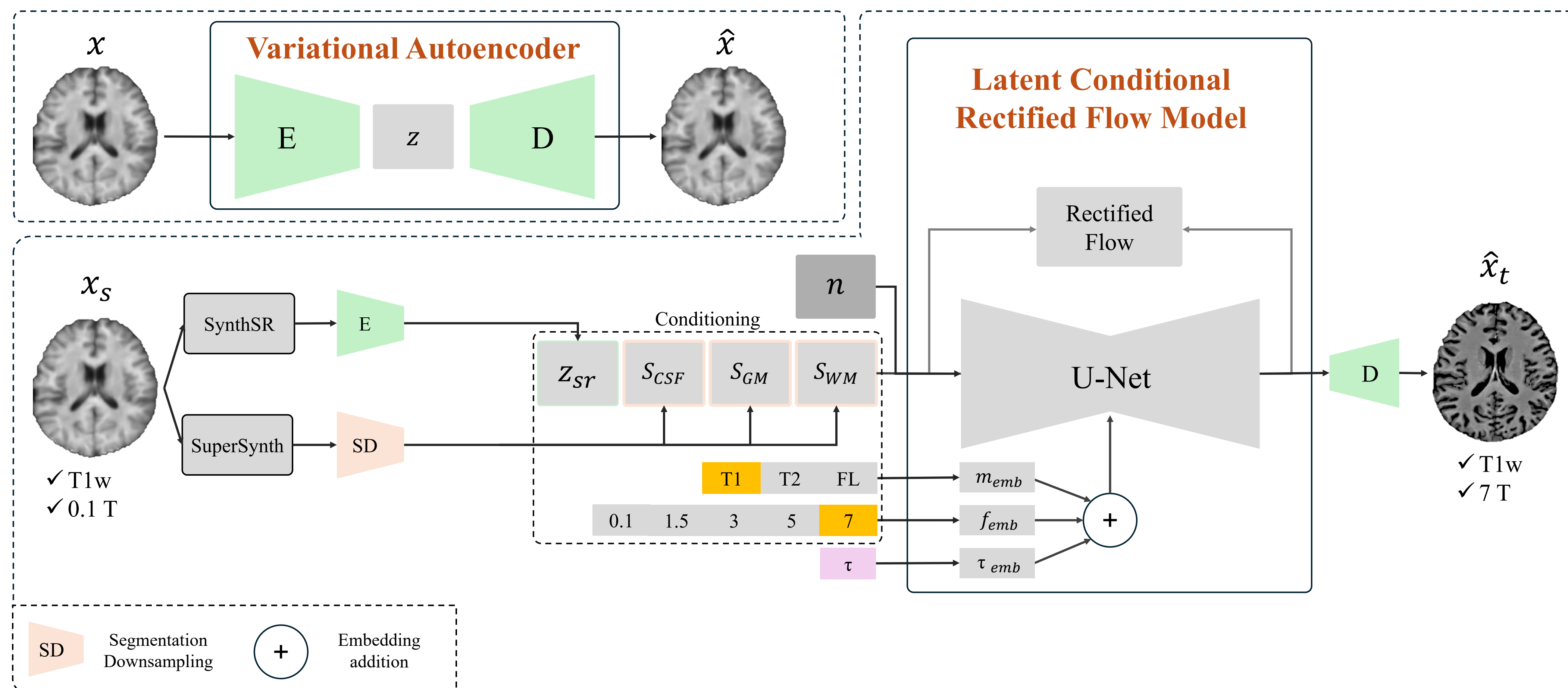


Figure 1. Overview of the proposed framework for arbitrary field-strength MRI translation, illustrated with 0.1T-to-7T T1w MRI synthesis. Anatomical information from SynthSR and SuperSynth is combined with Gaussian noise and provided to the diffusion model, together with the target modality and field-strength information. The model generates the target latent representation, which is decoded by the VAE to produce the final MRI.

HIGHLIGHTS

- **No paired data required.** Trained entirely on unpaired scans from multiple datasets [3–6], avoiding scarce and costly matched acquisitions.
- **Anatomy-guided generation.** Uses tissue maps (GM, WM, CSF) from segmentation [7], and a standardised reference image from SynthSR [8] to preserve anatomical structure.
- **Perturbed guidance during training.** Target-derived guidance is perturbed during training, preparing the model to use source-derived priors at inference.

CONCLUSION

LCRFM offers a unified route toward MRI harmonization, maintaining anatomical accuracy even under severe scanner mismatches. This comes at some cost to pixel-level sharpness compared with task-specific baselines.

RESULTS

Qualitative Results

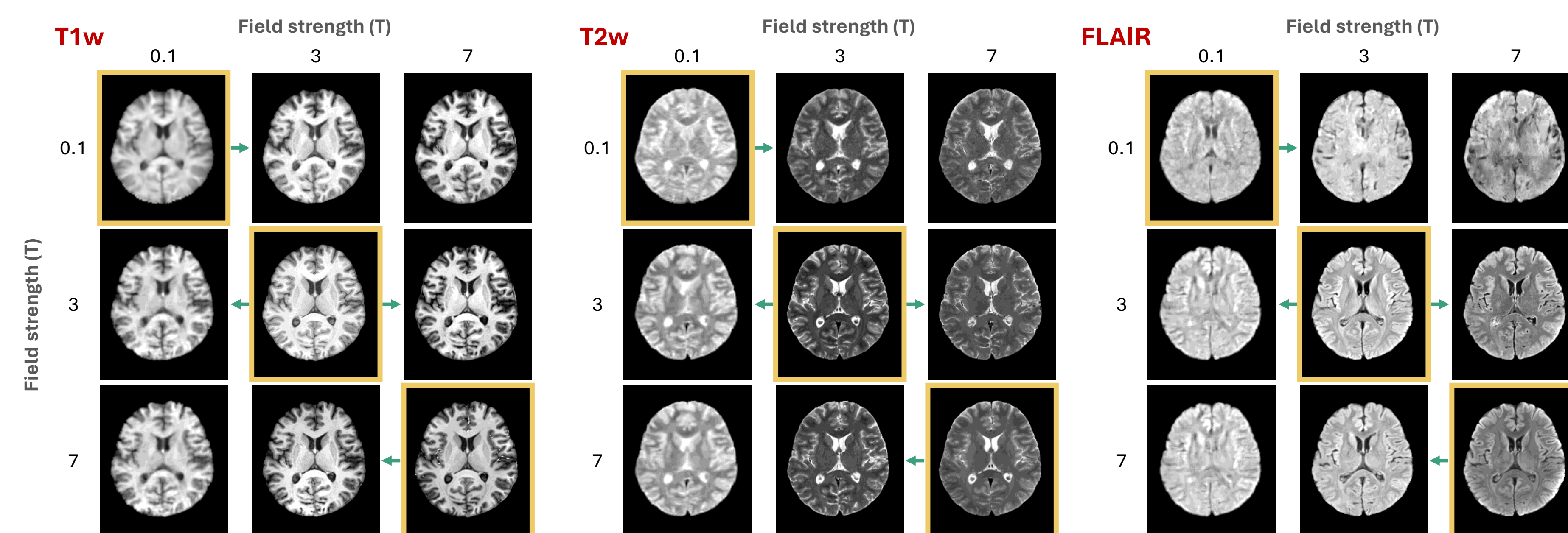


Figure 2. Field strength translation. Translation across field strengths (0.1T, 3T, 7T) for T1w, T2w, and FLAIR MRI. Yellow-bordered images indicate the source images, while arrows from the source indicate the synthesised images.

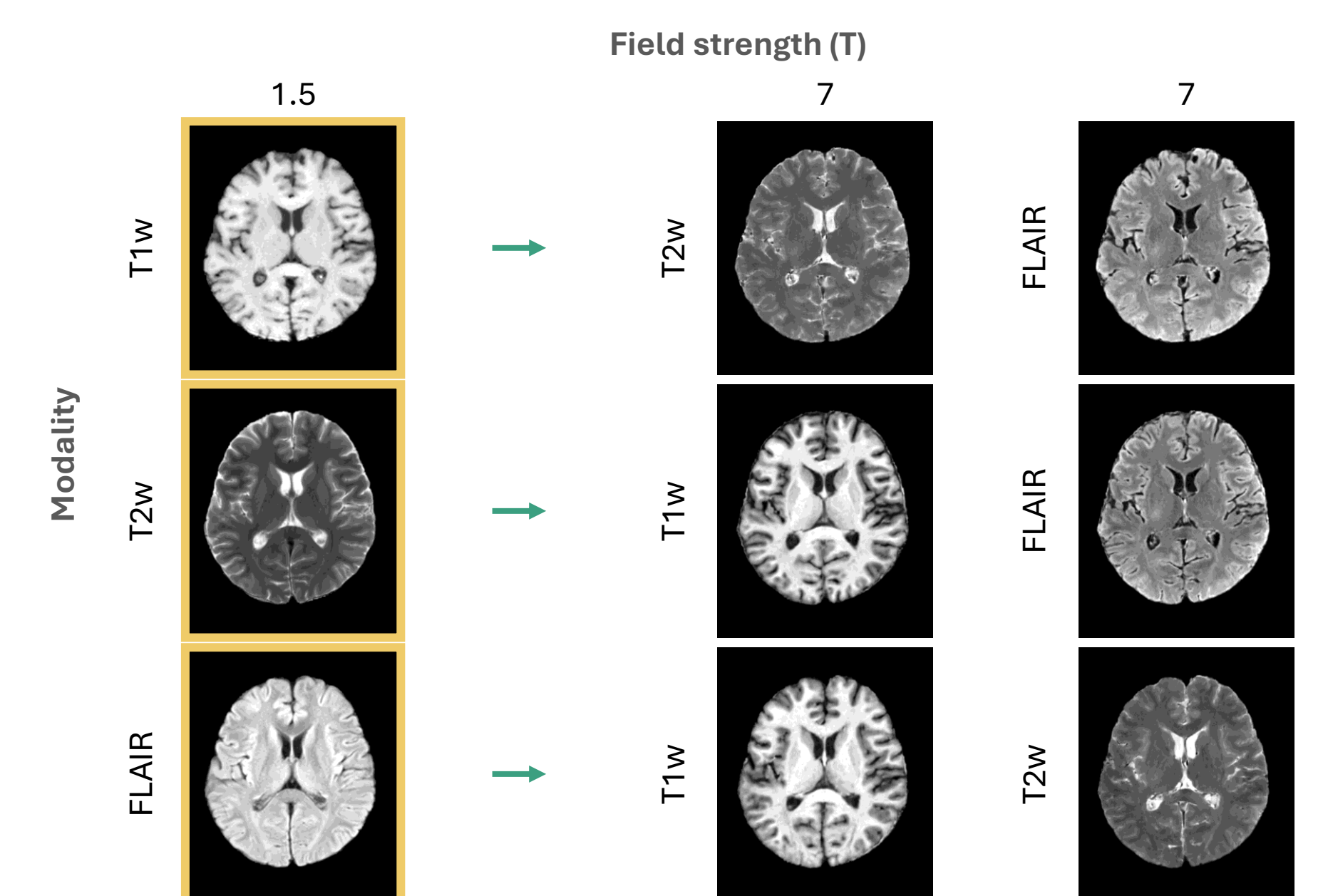


Figure 3. Multimodal translation. 1.5T-to-7T translation across T1w, T2w, and FLAIR, with each source modality translated to the other modalities.

Quantitative Results

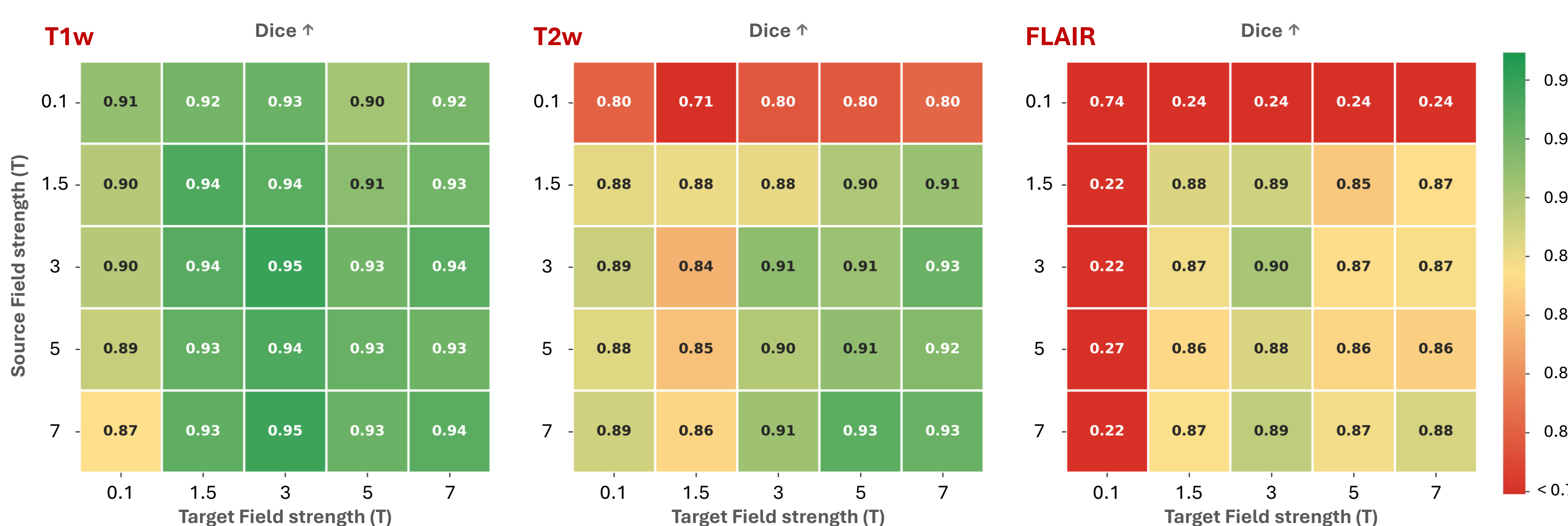


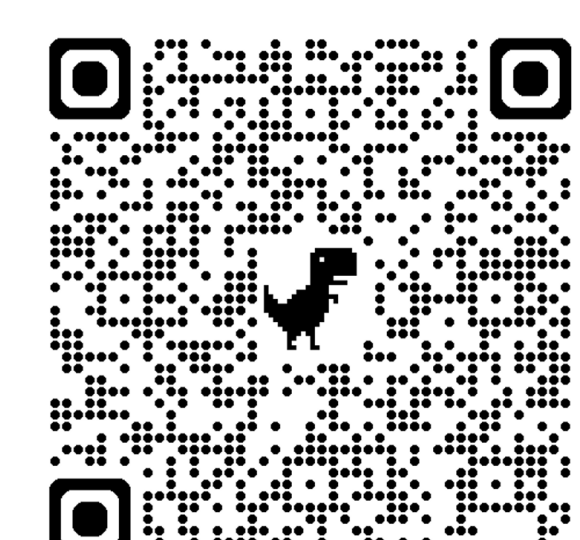
Figure 4. Mean Dice across source–target field-strength combinations for T1w, T2w, and FLAIR. Mean over 3 paired subjects; rows and columns denote source and target field strengths, respectively.

Task	Mod.	SSIM ↑	nRMSE ↓	LPIS ↓	Dice ↑	Volume ↑
Task 1	T1W	0.88 ± 0.04	0.29 ± 0.25	0.10 ± 0.02	0.85 ± 0.05	0.83 ± 0.13
	T2W	0.85 ± 0.03	0.27 ± 0.05	0.13 ± 0.02	0.86 ± 0.04	0.87 ± 0.05
	FLAIR	0.83 ± 0.03	0.38 ± 0.21	0.13 ± 0.03	0.85 ± 0.04	0.85 ± 0.10
Task 2	T1W	0.83 ± 0.03	0.34 ± 0.15	0.13 ± 0.02	0.84 ± 0.04	0.86 ± 0.05
	T2W	0.81 ± 0.02	0.31 ± 0.06	0.15 ± 0.02	0.81 ± 0.04	0.81 ± 0.06
	FLAIR	0.78 ± 0.03	0.30 ± 0.18	0.16 ± 0.02	0.80 ± 0.04	0.80 ± 0.09
Task 3	T1W	0.85 ± 0.19	0.32 ± 0.04	0.11 ± 0.02	–	–
	T2W	0.85 ± 0.04	0.30 ± 0.23	0.13 ± 0.03	–	–
	FLAIR	0.83 ± 0.03	0.24 ± 0.13	0.13 ± 0.03	–	–

Table 1. LCRFM per-modality performance on the validation set. Mean ± std across the three MRlxFIELDS2026 tasks, as reported on the official leaderboard. Dice/Volume are not part of the official Task 3 metrics.

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References, GitHub
code and PDF
version

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GITHUB CODE

<https://github.com/NIC-VICOROB/mrixfields2026-LCRFM>

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